

CUSTOMER BEHAVIOR ANALYSIS

Internship Task 1

Submitted By

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Objective

The objective of this project is to analyze customer transaction data to identify customer segments, understand purchasing behavior, analyze payment preferences, and evaluate customer churn risk

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab
- GitHub

Dataset Information

- Dataset Name: E-commerce Customer Data
- Total Records: 250,000
- Total Features: 13

Project Links

GitHub Repository:

<https://github.com/ramzeensarah-lgtm/Customer-Behavior-Analysis>

Google Colab Notebook:

[https://colab.research.google.com/github/ramzeensarah-lgtm/Customer-Behavior-Analysis/blob/main/Customer Behavior Analysis.ipynb](https://colab.research.google.com/github/ramzeensarah-lgtm/Customer-Behavior-Analysis/blob/main/Customer%20Behavior%20Analysis.ipynb)

Executive Summary

Customer behavior analysis was performed using an e-commerce transaction dataset. Data cleaning, feature engineering, customer segmentation, purchase pattern analysis,

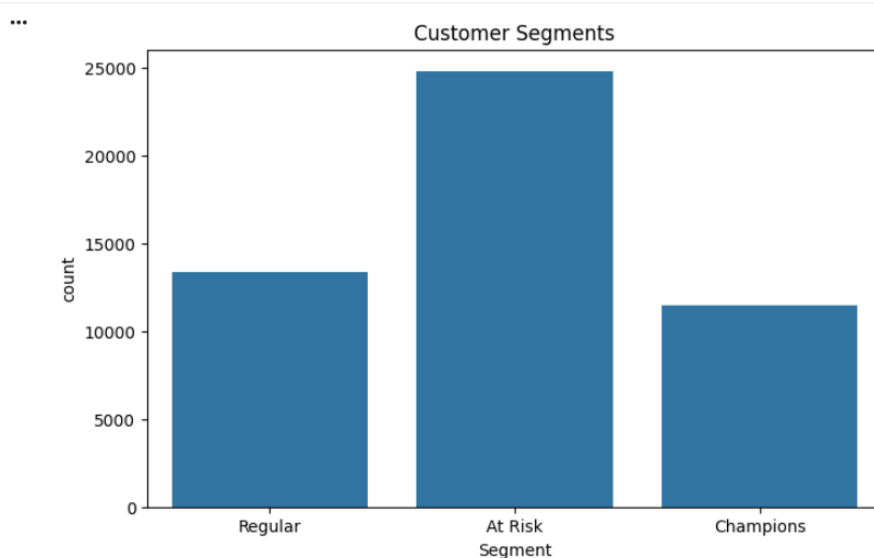
and churn analysis were conducted. RFM analysis was used to classify customers into Champions, Regular, and At Risk segments. The project provides insights that can help businesses improve customer retention and marketing strategies.

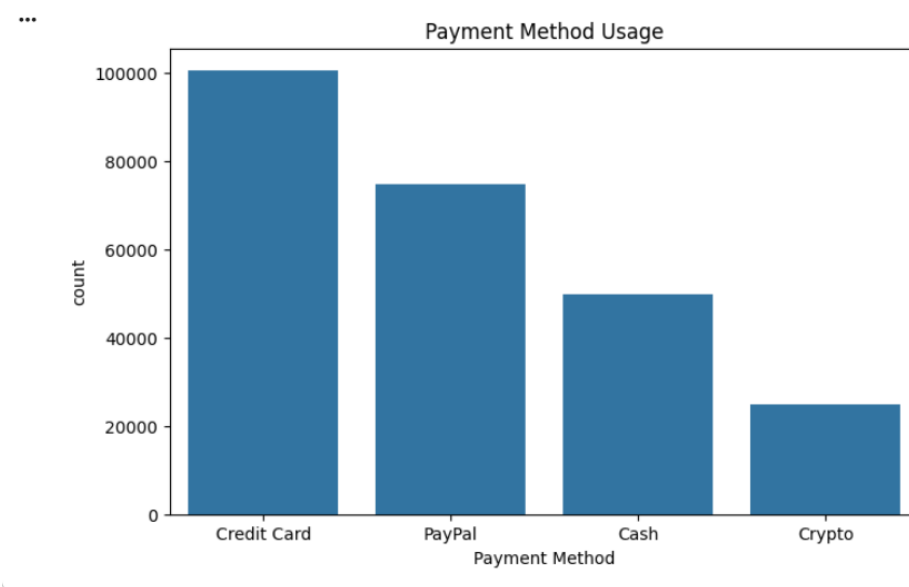
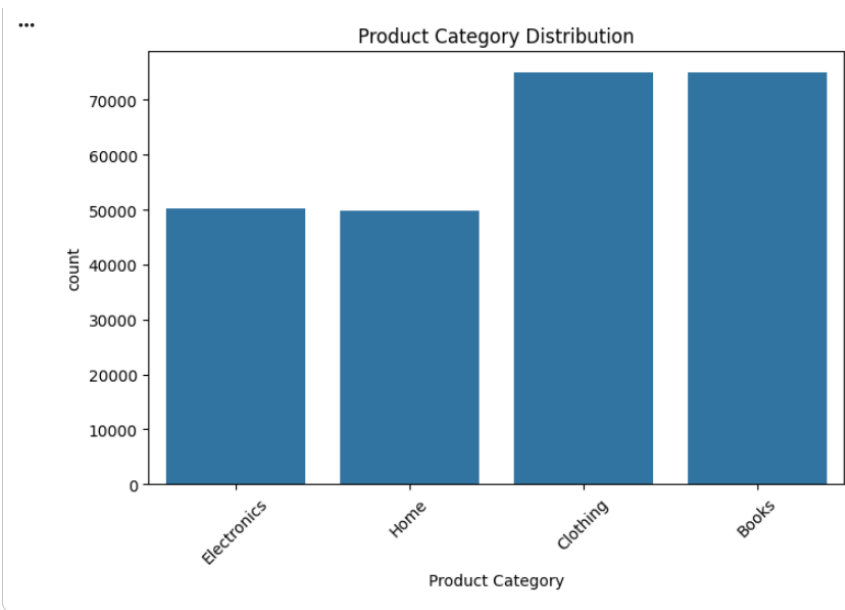
Key Findings

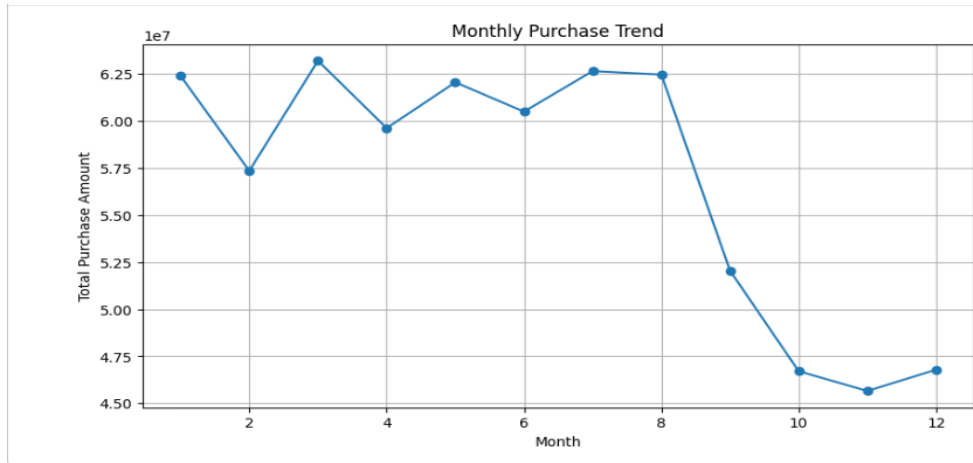
1. Total customers analyzed: 49,673
2. Champions customers: 11,466
3. Regular customers: 13,392
4. At Risk customers: 24,815
5. Clothing and Books were the most purchased categories.
6. Credit Card was the most preferred payment method.
7. Approximately 20% of customers had churned.

Recommendations

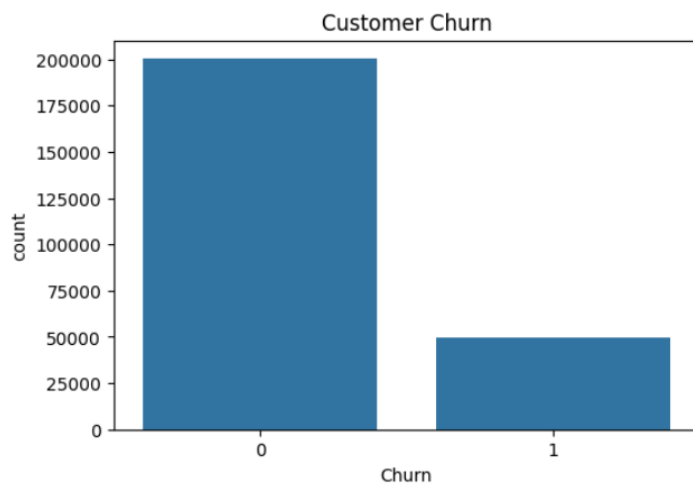
1. Reward Champion customers through loyalty programs.
2. Target At Risk customers with personalized offers.
3. Increase promotion of popular product categories.
4. Improve customer retention strategies.
5. Conduct seasonal marketing campaigns to boost sales.







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Conclusion

The project successfully analyzed customer behavior using transaction data. Customer segmentation identified Champions, Regular, and At Risk customers. Purchase pattern analysis revealed that Clothing and Books were the most popular product categories, while Credit Card was the most preferred payment method. Churn analysis showed that nearly 20% of customers had stopped purchasing. These insights can help businesses improve customer engagement, retention, and overall performance.